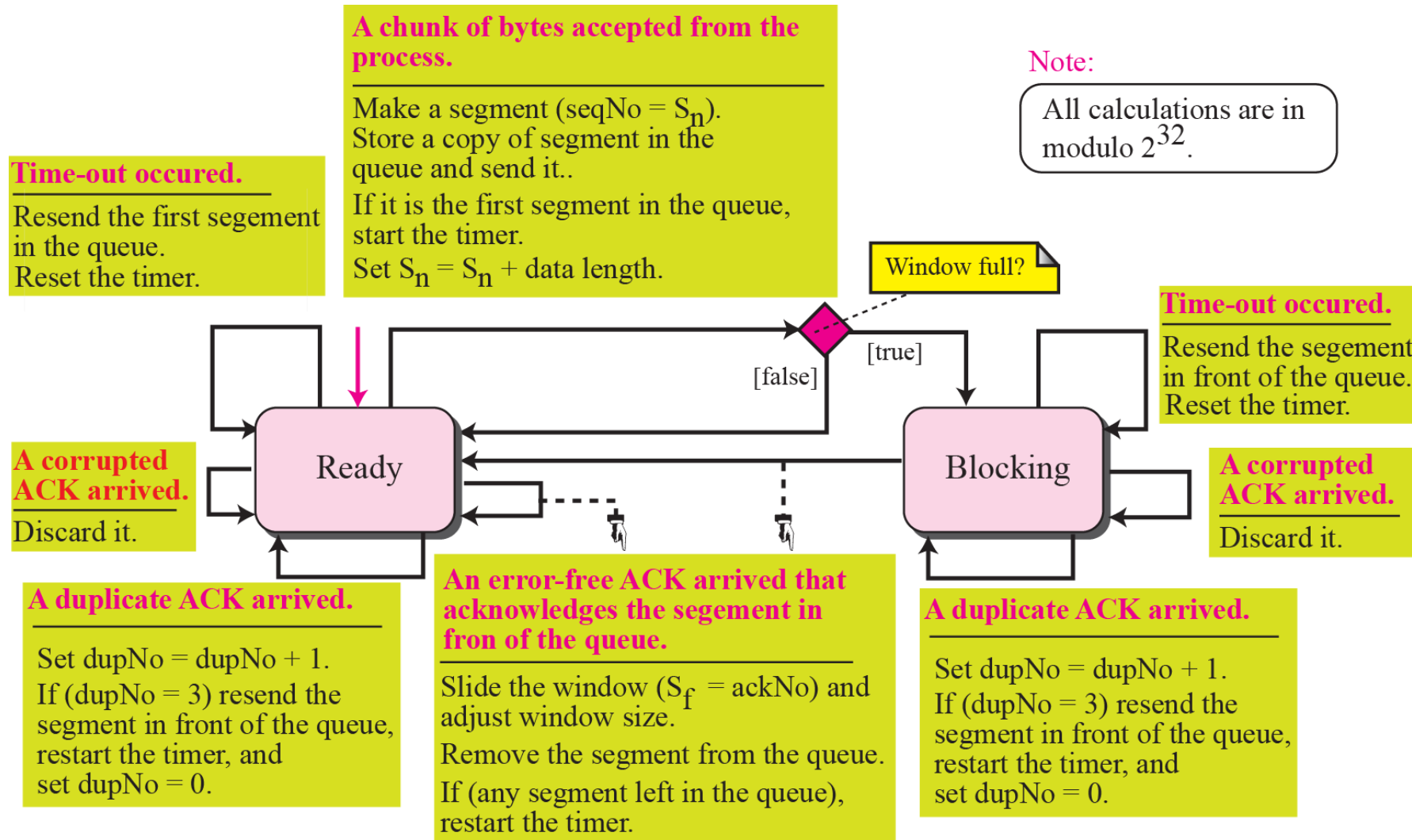


Transport Layer



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Simplified FSM for sender site



Simplified FSM for the receiver site

Note:

All calculations are in modulo 2^{32} .

An expected error-free segment arrived.

Buffer the message.

$R_n = R_n + \text{data length}$.

If the ACK-delaying timer is running, stop the timer and send a cumulative ACK. Else, start the ACK-delaying timer.

A request for delivery of k bytes of data from process came

Deliver the data.

Slide the window and adjust window size.

Ready

ACK-delaying timer expired.

Send the delayed ACK.

An error-free, but out-of order segment arrived

Store the segment if not duplicate.

Send an ACK with ackNo equal to the sequence number of expected segment (duplicate ACK).

An error-free duplicate segment or an error-free segment with sequence number outside window arrived

Discard the segment.

Send an ACK with ackNo equal to the sequence number of expected segment (duplicate ACK).

A corrupted segment arrived

Discard the segment.

TCP round trip time, timeout

Q: how to set TCP timeout value?

- longer than RTT, but RTT varies!
- *too short*: premature timeout, unnecessary retransmissions
- *too long*: slow reaction to segment loss

Q: how to estimate RTT?

- *SampleRTT*: measured time from segment transmission until ACK receipt
 - ignore retransmissions
- *SampleRTT* will vary, want estimated RTT “smoother”
 - average several *recent* measurements, not just current *SampleRTT*